



Research progress on the toxicity of toxic Traditional Herbals from Thymelaeaceae

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ABSTRACT

Ethnopharmacological relevance: Plants from the Thymelaeaceae family are widely distributed in tropical and temperate regions, with approximately 113 species used as Traditional Herbals. There are numerous applications for them, such as treating leukemia, AIDS, and liver cancer. It should be noted that around 20% of these plants have shown harmful side effects when used in clinical applications, including solid irritations to the skin and mucous membranes, carcinogenic effects, organ damage, vomiting, and diarrhea.

Aim of the study: This paper aims to review the toxic side effects, toxic compounds, toxic mechanisms, and detoxification methods of Traditional Herbals in Thymelaeaceae, guiding their safe clinical uses.

Materials and methods: This review employed the keywords "Thymelaeaceae," 48 different "genus," 966 "species," and the combination of "toxicity" to identify the medicinal value and toxicity of plants from Thymelaeaceae in scientific databases (Pubmed, SciFinder Scholar, Elsevier, Web of Science, and CNKI). Information relevant to the toxicity of Traditional Herbals from Thymelaeaceae up to June 2023 has been summarized. The plant names have been checked with "World Flora Online" (www.worldfloraonline.org).

Results: 28 toxic Traditional Herbals from 13 genera within the Thymelaeaceae family were categorized. Toxicities were summarized at the cellular, animal, and clinical levels. The toxic substances are primarily concentrated in the *Daphne* L. and *Wikstroemia* Endl. genera, with terpenes being the main toxic components. The toxicity mechanism is primarily associated with the mitochondrial pathways. Detoxification and enhanced efficacy can be achieved through processing methods such as vinegar-processing and sweat-soaking.

Conclusions: Medicinal plants in the Thymelaeaceae exhibit significant pharmacological activities, such as anti-HIV and anti-tumor effects, indicating a broad potential for application. However, their clinical uses are hindered by their inherent toxicity. Researching the toxic components and mechanisms of these Traditional Herbals and exploring more effective detoxification methods can contribute to unveiling the latent value of these medicinal plants from Thymelaeaceae.

1. Introduction

The Thymelaeaceae family comprises over 966 species in 48 genera and 113 species have been used as Traditional Herbals. Among them, 28 species exhibit explicit toxicity. For instance, *Daphne genkwa* Siebold & Zucc. is categorized as a "lower-grade" herb due to its significant toxicity in the classic Chinese medicinal text "Shennong Ben Cao Jing." It is used clinically for diuresis, resolving water retention, and expelling phlegm to alleviate cough. Prolonged or excessive consumption can lead to hemolysis and damage to internal organs such as the heart, kidneys, and liver. In "Compendium of Materia Medica," it is recorded that "*Stellera*

chamaejasme has great toxicity; its name suggests its poison." *S. chamaejasme* L. is known for its diuretic, accumulations-resolving, and insecticidal properties. Excessive consumption can lead to acute poisoning symptoms such as vomiting, severe abdominal pain, and other adverse effects. *Wikstroemia indica* (L.) C.A.Mey. is employed for heat-clearing, detoxification, reducing swelling, and dispersing accumulations. Clinical usage can lead to gastrointestinal side effects such as nausea, vomiting, and upper abdominal discomfort.

Existing literature provided comprehensive reports on medicinal plants' chemical composition and pharmacological activities within the Thymelaeaceae family. However, a comprehensive and in-depth

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summary of their toxicity remains unavailable. This article aims to review the toxic side effects, toxic compounds, toxic mechanisms, and detoxification methods of toxic Traditional Herbals from Thymelaeaceae. The intention is to offer guidance for the research, development, and safe clinical application of the medicinal plants from Thymelaeaceae.

2. Toxic Traditional Herbals from thymelaeaceae

Toxic Traditional Herbals within the Thymelaeaceae family are distributed among 28 species across 13 genera, with 46% originating from the *Daphne* and *Wikstroemia* genera. Their medicinal parts, distribution, ethnopharmacological uses, and toxicity are detailed in Table 1.

3. The toxic components of toxic Traditional Herbals from thymelaeaceae

Currently, there are 60 toxic components reported from toxic Traditional Herbals in the Thymelaeaceae family, including 34 terpenoids, 15 flavonoids, and 11 other types of compounds. Diterpenoids and flavonoids are the main types of toxic components, some of which exhibit bioactive properties. For instance, daphnane-type and tiglane-type diterpenoid esters manifest the traditional Chinese medicine principle of "using poison to counteract poison," applying the concept of "selectivity" to balance imbalances and treat diseases with herbal medicine (Hayes et al., 2010).

3.1. Terpenoids

34 toxic terpenoids (1–34) have been reported in toxic Traditional Herbals from Thymelaeaceae, including 22 diterpenoids, 8 sesquiterpenoids, and 4 triterpenoids. The structures of these toxic terpenoid compounds are shown in Fig. 1, and their names, plant sources, and references are presented in Table 2.

3.2. Flavonoids

15 toxic flavonoid compounds (35–49) have been reported in toxic Traditional Herbals from the Thymelaeaceae family. These include 9 simple flavonoid compounds and 6 biflavonoid compounds, primarily found in *S. chamaejasme*. The structures of these toxic flavonoid compounds are illustrated in Fig. 2, and their names, plant sources, and references are presented in Table 3.

3.3. Others

11 other toxic components (50–60) have been reported in toxic Traditional Herbals from the Thymelaeaceae family. The structures of these other types of toxic components are illustrated in Fig. 3, and their names, plant sources, and references are presented in Table 4.

4. The toxicity of toxic Traditional Herbals from thymelaeaceae

The toxicity of toxic Traditional Herbals from Thymelaeaceae can be categorized based on different levels of research: cellular level, animal level, and clinical application. Their toxicity to animals and humans is presented in Fig. 4.

4.1. Cytotoxicity

The ethanol extract of *S. chamaejasme* exhibited a dose-dependent inhibition of the growth and proliferation of Wistar rat lung fibroblasts. The extract contained the flavonoid compound isochamaejasmin (ICM, 44), which induced toxic reactions in corn earworm (*Helicoverpa zea*) AW1 neurons, including DNA damage, depolarization of mitochondrial membrane potential, dose-dependent elevation of reactive

oxygen species (ROS) levels, accumulation of lipid peroxidation products, and deactivation of antioxidant enzymes (Ren et al., 2021). Neochamaejasmin B (NCB, 45) caused a dose-dependent reduction in AW1 cell viability. Transmission electron microscopy revealed the disappearance of the nuclear membrane and the appearance of apoptotic bodies in AW1 cells treated with NCB, indicating morphological changes associated with apoptosis (Gu et al., 2022).

The ethanol extract of *W. indica* exhibited toxicity on Vero cells, with the petroleum ether, ethyl acetate, and n-butanol fractions all showing toxic effects. The toxicity increased as the polarity of the extraction solvent decreased: petroleum ether extract (CC₅₀: 3.12 µg/mL) > ethyl acetate extract (CC₅₀: 46.87 µg/mL) > n-butanol extract (CC₅₀: >250.00 µg/mL) (Xue, 2007).

The extract from *D. genkwa* and its diterpenoid ester component, yuanhuacin (9), exhibited significant cytotoxic effects on normal human liver cells (L-02). The IC₅₀ values after 48 h of exposure were 48.34 ± 4.66 and 29.57 ± 2.01 mg/L, respectively (Shi et al., 2013). A series of diterpenoid compounds, including genkwanine D, genkwanine J, yuanhuacin, and yuanhuadin (23, 27, 9, 10), isolated from the buds of *D. genkwa* demonstrated significant inhibitory activity on the proliferation of human microvascular endothelial cells. The IC₅₀ values for these compounds were 2.90, 8.10, 14.00, and 9.00 µM, respectively (Zhan et al., 2005).

The dirchomone compounds (51–55) from the dichloromethane extract of *D. palustris* exhibited toxicity to human normal skin fibroblasts, with IC₅₀ values of approximately 9 µM, 6 µM, and 9 µM for dirchomone, 6-methoxydirchomone, and 7-methoxydirchomone, respectively (St-Gelais et al., 2015).

The daphnetoxin (20), a daphnane-type diterpenoid widely present in plants of the Thymelaeaceae family, and isodaphnoretin (50) were toxic to the liver mitochondria of Wistar rats. The toxicity was most pronounced when the concentration of daphnetoxin was higher than 100 µM. The toxicity manifested as an increase in inner mitochondrial membrane proton leakage, augmentation of mitochondrial permeability transition pore, inhibition of ATP synthase, and suppression of the mitochondrial respiratory chain (Diogo et al., 2009).

4.2. In vivo toxicity in animals

4.2.1. *Daphne genkwa* Siebold & Zucc

Long-term in vitro exposure to the flower buds of *D. genkwa* resulted in motor neuron dysfunction, disruption of intestinal development, and alterations in defecation behavior in *Caenorhabditis elegans* (Qiao et al., 2014). Rats that consumed the flower buds of *D. genkwa* displayed liver sinusoidal congestion, hepatocyte swelling, and with noticeable glomerular enlargement and renal tubular degeneration (Jiang et al., 2014). Zebrafish treated with aqueous extract of *D. genkwa* exhibited pathological changes in liver tissue, including hepatocyte atrophy, deformation, vacuolization, blurred boundaries, and lipid accumulation (Han et al., 2020). Mice orally administered yuanhuapine (22) at a dose of 10.0 mg/kg experienced respiratory distress, abdominal breathing symptoms, and significant seizures. Intraperitoneal injection of yuanhuapine resulted in an LD₅₀ of 1.40 mg/kg (Chen et al., 2015; Hu et al., 1986). Intraperitoneal injection of yuanhuacin (9) at a dose of 1 mg/kg led to a notable reduction in body weight in mice and occasional fatality (Bailey, 2022).

4.2.2. *Stellera chamaejasme* L

After gastric administration of water extract from *S. chamaejasme* to Plateau zokors (*Eospalax baileyi*), liver granular degeneration and hepatic inflammatory responses occurred, accompanied by a dose-dependent increase in serum levels of aspartate aminotransferase, alanine aminotransferase, and malondialdehyde (Tan et al., 2022). The root's petroleum ether extract demonstrated strong lethality against *Fagopyrum esculentum*, with a 24-h LC₅₀ of 1243.7 mg/L (Gao et al., 2022). Adult sheep that consumed the powder of *S. chamaejasme*

Table 1

Toxic traditional herbals from thymelaeaceae.

No.	Species	Part of plants	Country/Region	Ethnopharmacological applications	Toxicity	References
1	<i>Daphne genkwa</i> Siebold & Zucc.	Flower buds	China, Korea	Treatment of chronic bronchitis, hepatitis, edema, asthma, leukemia, phlegm expulsion, cough relief, mid-term pregnancy termination, immune function regulation	Skin and mucous membrane irritations, hepatotoxicity, nephrotoxicity, gastrointestinal toxicity, insecticidal effects	Chen et al. (2013), Han et al. (2020), Li et al. (2007), Wang and Cao (1999), Xiang et al. (2006)
2	<i>Stellera chamaejasme</i> L.	Roots	China, Russia	Treatment of AIDS, chronic bronchitis, pulmonary tuberculosis, scabies, neurodermatitis, hyperplastic scars, phlegm expulsion, cough relief, and alleviation of abdominal distention	Skin and mucous membrane irritations Gastrointestinal symptoms: vomiting, diarrhea Neurological symptoms: insomnia, spasms, insecticidal effects	Gao et al. (2022), Han et al. (2013), He et al. (2006), Ren et al. (2019), Zhang et al. (2021)
3	<i>Wikstroemia indica</i> (L.) C.A.Mey.	Roots, root bark	China, Vietnam, India	Treatment of contusions, parotitis, rheumatoid arthritis, hepatitis, nephritis, pneumonia, bronchitis, gynecological inflammation	Skin and mucous membrane irritations Gastrointestinal symptoms: vomiting, diarrhea, insecticidal effects	Xue (2007), Zheng and Feng (2019)
4	<i>Wikstroemia chamaedaphne</i> Meisn.	Flowers, leaves, seeds, roots, root bark	China, Russia, Japan, Poland	Treatment of edema, acute and chronic hepatitis, schizophrenia, epilepsy, induced abortion, phlegm expulsion, cough relief	Skin and mucous membrane irritations, embryotoxicity, insecticidal effects	Zhao et al. (2004)
5	<i>Wikstroemia retusa</i> A.Gray	Bark	Taiwan, Philippines, Japan, Italy	Treatment of AIDS, leukemia, hair care	Insecticidal effects	Sogabe et al. (2000)
6	<i>Wikstroemia canescens</i> Meisn.	Roots	China	Treatment of tumors	Skin irritation	Dagang et al. (1993)
7	<i>Daphne giraldii</i> Nitsche	Stem bark, root bark	China	Relaxing tendons and bones, reducing swelling, relieving pain	Fever, drowsiness, skin irritation: skin burning sensation, blistering, and rash	Gong and Li (2015)
8	<i>Daphne papyracea</i> Wall. ex G.Don	Roots, stem bark	China, Kashmir, India, Nepal, Bhutan, Sikkim	Treatment of hypertension	Disappearance of righting reflex, paralysis, and even death	Gong and Li (2015), Raina et al. (2014)
9	<i>Daphne kiusiana</i> Miq.	Roots, stem bark	China	Treatment of rheumatic pain, bruises and injuries, sore throat, toothache, and skin infections	Disappearance of righting reflex and even death	Gong and Li (2015), Jia and Min (2005)
10	<i>Daphne mezereum</i> L.	Flowers, branches	Russia, Norway, Italy	Treatment of tumors: inhibits the proliferation of human cancer cells MCF-7, A549, LNCaP, ACHN, and C32	Skin irritation: rash, gastrointestinal symptoms	Gorick and Melzig (2013), Tundis et al. (2019)
11	<i>Daphne oleoides</i> Schreb.	Leaves, roots	India, Italy, Turkey	Treatment of inflammation, ascites, toothache, skin disorders, epilepsy, wound healing, and as a laxative	Skin irritation	Rahman and Jahan (2021), Rahman et al. (2020)
12	<i>Daphne tangutica</i> Maxim.	Stem bark, root bark	China	Treatment of rheumatic pain, rheumatoid and similar arthritic conditions, contusions, and as an anti-inflammatory and analgesic	Insecticidal effects	Li et al. (2011), Xu et al. (2000)
13	<i>Daphne aurantiaca</i> Diels	Bark, stem bark, roots	China	Treatment of AIDS, contusions and injuries, and anti-inflammatory purposes	Insecticidal effects	Huang et al. (2017, 2015, 2014), Liang et al. (2010)
14	<i>Daphne mucronate</i> Royle	Whole plant	Iran, Pakistan	Inhibition of liver and kidney toxicity caused by antipyretic analgesics and treatment of rheumatism, gonorrhea, osteoporosis, facial freckles, eye issues	Insecticidal effects	Ashraf et al. (2018), Khan et al. (2017), Nazir et al. (2021)
15	<i>Dirca palustris</i> L.	Bark, branches	Canada, the United States	Inducing vomit, promoting bowel movement, and exhibiting anticancer properties	Skin irritation: redness, vascular occlusion, ulcers Gastrointestinal symptoms: vomiting, diarrhea	Ramsewak et al. (2001), St-Gelais et al. (2015)
16	<i>Phaleria macrocarpa</i> Boerl.	Fruit peel	Malaysia, Indonesia, New Guinea	Treatment of hypertension, diabetes, high cholesterol levels, ulcers, skin disorders, and cancer combining with other drugs	Insecticidal effects	Abood et al. (2015), Kurnia et al. (2008), Lay et al. (2014)
17	<i>Gyrinops versteegii</i> (Gilg) Domke	Leaves	Indonesia	Treatment of diabetes and as an antioxidant	Insecticidal effects	Sukito et al. (2020)
18	<i>Linostoma decandrum</i> (Roxb.) Steud.	Roots	India	- ^a	Insecticidal effects	Dohutia et al. (2015)
19	<i>Ovidia andina</i> Meisn.	- ^a	Argentina, Chile	- ^a	Insecticidal effects	Tarcaya et al. (2014)

(continued on next page)

Table 1 (continued)

No.	Species	Part of plants	Country/Region	Ethnopharmacological applications	Toxicity	References
20	<i>Phaleria nisidai</i> Kaneh.	Leaves	Palau	Treatment of diabetes, common illnesses, and influenza; enhancing immunity, and reducing blood pressure	Diarrhea	Kitalong et al. (2017), Kulakowski et al. (2015, 2014)
21	<i>Pimelea elongate</i> Threlfall	Leaves, roots	Australia	Providing nerve nourishment and exhibiting anti-tumor effects	Subcutaneous edema, diarrhea, anemia, heart failure	Hayes et al. (2009, 2010)
22	<i>Pimelea prostrata</i> Willd.	Leaves, roots	New Zealand, Australia	Anti-tumor effects	Pro-inflammatory, carcinogen	Zayed et al. (1982, 1977)
23	<i>Pimelea simplex</i> F. Muell.	Stems	Australia	Treatment of leukemia	Edema, anemia, chronic diarrhea, circulatory failure, hepatic sinusoid dilatation	Chow et al. (2010), McClure and Mulley (1984), Tyler and Howden (1985)
24	<i>Struthiola argentea</i> Lehm.	Whole plant	South Africa	- ^a	Insecticidal effects	Ayers et al. (2008)
25	<i>Synaptolepis kirkii</i> Oliv.	Roots	Kenya, Tanzania	Treatment of epilepsy, snakebite envenomation, leukemia; neuro-nutrition	Skin irritation, vomiting	Adolf et al. (1988), He et al. (2002, 2000)
26	<i>Synaptolepis retusa</i> H.Pearson	Roots	West Africa	Antitumor	Skin irritation	Adolf et al. (1988)
27	<i>Lasiosiphon kraussianus</i> (Meisn.) Hutch. & Dalziel	Roots	Guinea, Nigeria, Mozambique, Sudan	Treatment of leukemia	Insecticidal effects	Bala et al. (1999)
28	<i>Lasiosiphon latifolius</i> (Oliv.) Brenan	Leaves	Kenya, India, Madagascar	Treatment of leukemia	Acute poisoning: gastrointestinal mucosal bleeding Chronic poisoning: generalized emaciation, body cavity effusion	Kiptoon et al. (1982)

^a No results found.

exhibited symptoms, including digestive system disruption, gastric-intestinal tissue damage, and diarrhea (Shen et al., 2015). Chang et al. (2002) conducted an in vivo toxicity assessment of flavonoid compounds present in *S. chamaejasme* using rabbits. Rabbits fed with 800.0 mg/kg of flavonoids from *S. chamaejasme* exhibited varying degrees of fatigue, diarrhea, respiratory failure, visceral damage, and even death between the 6th and 10th days. Gao et al. isolated 1,5-diphenyl-1-pentanone (57) and 1,5-diphenyl-2-penten-1-one (58) from *S. chamaejasme*, which demonstrated strong contact irritancy and antifeedant activity against *Aphis gossypii* and *Schizaphis graminum*. After 24 h of contact, the LD₅₀ values for *A. gossypii* were 469 mg/L and 232 mg/L, respectively, for the two compounds, and for *S. graminum*, the LD₅₀ values were 443 mg/L and 195 mg/L (Gao et al., 2001). The compound ICM (44) in *S. chamaejasme* exhibited insecticidal properties against *H. zea* in a time- and dose-dependent manner (Ren et al., 2021). Furthermore, 1,5-diphenyl-3-hydroxyl-1-pentanone (56) inhibited acetylcholinesterase activity in *Pieris rapae*, resulting in antifeedant effects (He et al., 2006).

4.2.3. *Wikstroemia indica* (L.) C.A.Mey

The ethyl acetate extract of *W. indica* demonstrated significant hepatotoxicity, as evidenced by notable elevation in serum hepatic biochemical markers, including ALP, T-BIL, and GLU in administered rats, along with a significant decrease in A/G, TP, and ALB levels ($P < 0.05$) (Feng et al., 2018).

4.2.4. *Daphne aurantiaca* diels

Due to its insecticidal activity, *D. aurantiaca* is used in Tibetan Buddhism to make insect-repellent paper. Huang et al. isolated three sesquiterpenoids, daphnauranols A-C (1–3), from the stems of *D. aurantiaca*, which exhibited antifeedant activity against fruit flies with antifeedant indices (AI) of $39.8 \pm 7.2\%$, $29.4 \pm 7.2\%$, and $26.3 \pm 6.4\%$, respectively (Huang et al., 2014). Five sesquiterpenoids, daphnauranins A-E (6, 8, 4, 5, 7), isolated from the roots of *D. aurantiaca*, showed antifeedant activity against male fruit flies with AI values of $46.2 \pm 7.1\%$, $44.7 \pm 5.4\%$, $39.1 \pm 5.8\%$, $39.2 \pm 3.9\%$, and $27.8\% \pm 6.2\%$, respectively (Huang et al., 2017, 2018).

4.2.5. Other toxic medicinal plants

Various medicinal plants within the Thymelaeaceae family exhibit potent insecticidal activity. For instance, root extracts of *L. kraussianus* and *L. decandrum* (Bala et al., 1999; Dohutia et al., 2015), as well as hexane extracts of *D. palustris* seeds (Ramsewak et al., 2001), have demonstrated notable insecticidal properties. Furthermore, methanol extracts of *D. tangutica* exhibited antifeedant and gastric toxicity against *P. rapae* larvae, leading to reduced larval size, disruption of midgut tissue cells, and decreased hemolymph protein content (Xu et al., 2000). Linoleic acid (59) and oleic acid (60) from *D. palustris* exhibited LD₅₀ values of 100 µg/mL against *Aedes aegyptii* larvae for 24 h. These two compounds also caused growth inhibition in insect species such as *H. zea*, *Lymantria dispar*, *Orgyia leucostigma*, and *Malacosoma disstria*, reducing growth by 88% and 85%, 93% and 91%, 81% and 80%, and 77% and 75%, respectively (Ramsewak et al., 2001).

The root extract of *S. retusa* exhibited tumor-promoting toxicity and irritant toxicity in mice (Adolf et al., 1988). The methanol extract of *W. canescens* root containing phorbol-12-benzoate-13-decadienoate (28) and prostratin (29) from *P. prostrata* both displayed irritant toxicity on mouse ears (Dagang et al., 1993; Zayed et al., 1977, 1982).

Cattle ingesting toxic plants such as *P. trichostachya*, *P. simplex*, and *P. elongata* exhibited symptoms of right ventricular dilation and circulatory failure, accompanied by edema, anemia, chronic diarrhea, and hepatic sinusoid dilation (Chow et al., 2010; McClure and Mulley, 1984; Tyler and Howden, 1985). Consumption of *L. latifolius* leaves led to elevated urea nitrogen and lactate dehydrogenase levels, accompanied by systemic emaciation, hepatic fibrosis, and submucosal hemorrhage in the gastrointestinal tract (Kiptoon et al., 1982). Intravenous injection of simplexin (12) resulted in a threefold increase in pulmonary artery pressure within a hundred seconds, while oral administration led to adverse effects such as anorexia, weight loss, jugular vein distention, and mandibular edema (McClure and Mulley, 1984).

4.3. Clinical toxic side effects

"The Compendium of Materia Medica" records that *S. chamaejasme* can cause acute poisoning symptoms, such as vomiting, diarrhea, limb

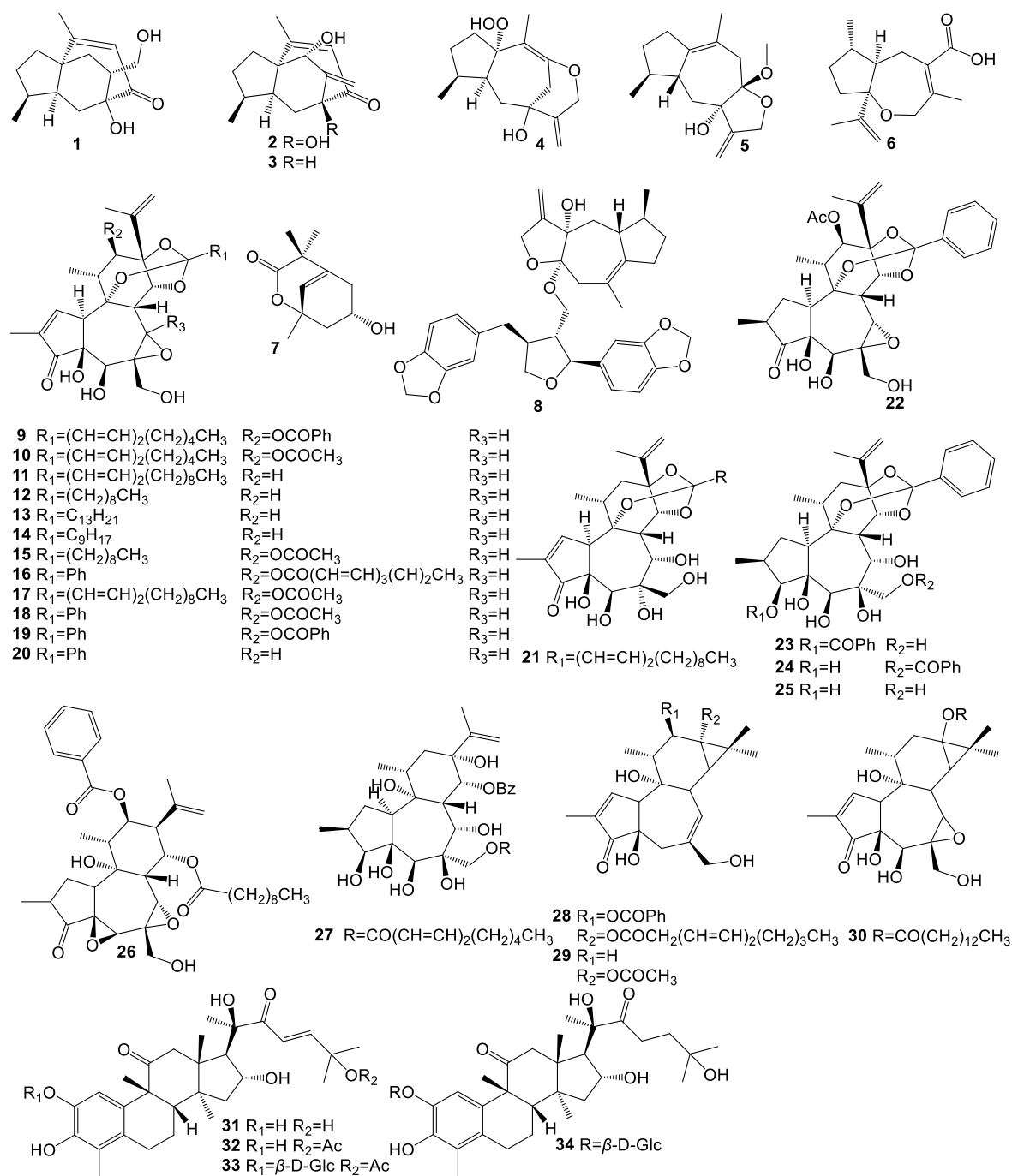


Fig. 1. Toxic terpenoid compounds in toxic Traditional Herbals from Thymelaeaceae.

weakness, and generalized spasms. In severe cases, it can lead to convulsions and death (Zhang et al., 2021). Improper external use of *S. chamaejasme* can lead to poisoning, manifesting as itching, blistering, and other symptoms. Contact of its sap with the eyes can lead to blindness, and oral ingestion can cause symptoms such as sore throat and gastrointestinal discomfort (Han et al., 2013).

W. indica is highly toxic, with its poisonous fruits, stems, leaves, and root bark. Poisoning from *W. indica* can lead to digestive system toxic reactions such as vomiting and diarrhea (Yi et al., 2012). An incident was reported in 2008 where severe vomiting, discoloration of extremities, difficulty breathing, and death occurred 4 h after ingesting 1.5 packets of *W. indica* powder (Zhang et al., 2008). In 2018, a case was reported involving severe erythema multiforme drug eruption with

drug-induced liver injury due to repeated consumption of *W. indica* tablets three times (3 tablets per time) following exposure to cold (Liu et al., 2018).

D. oleoides leaves have traditionally been used to treat inflammatory diseases. In Unani medicine, it is categorized as a fourth-degree drug due to its toxicity in all parts, and its usage is recommended only after detoxification. Skin contact with its sap can lead to dermatitis. Using untreated coarse leaves of *D. oleoides* can cause vomiting, diarrhea, and severe complications (Rahman and Jahan, 2021).

W. chamaedaphne, also known as 'Yang Yanyan' or 'Laohu Ma,' contains terpenoid compounds in its leaves that strongly irritate the skin and mucous membranes. Accidental contact of its preparations with the eyes can lead to symptoms such as conjunctival congestion, corneal

Table 2
Toxic terpenoid compounds in toxic Traditional Herbals from Thymelaeaceae.

No.	Compound	Species	References
1	daphnauranol A	<i>D. aurantiaca</i>	Huang et al. (2014)
2	daphnauranol B	<i>D. aurantiaca</i>	Huang et al. (2014)
3	daphnauranol C	<i>D. aurantiaca</i>	Huang et al. (2014)
4	daphnauranin C	<i>D. aurantiaca</i>	Huang et al. (2018)
5	daphnauranin D	<i>D. aurantiaca</i>	Huang et al. (2018)
6	daphnauranin A	<i>D. aurantiaca</i>	Huang et al. (2017)
7	daphnauranin E	<i>D. aurantiaca</i>	Huang et al. (2018)
8	daphnauranin B	<i>D. aurantiaca</i>	Huang et al. (2017)
9	yuanhuacin	<i>D. genkwa</i>	Chen et al. (2013), Zhan et al. (2005)
10	yuanhuadin	<i>D. genkwa</i>	Chen et al. (2013), Zhan et al. (2005)
11	huratoxin	<i>P. nisidai</i>	Kulakowski et al. (2015)
12	simplexin	<i>P. nisidai</i>	Kulakowski et al. (2015)
13	dehydrohuratoxin	<i>P. elongate, P. nisidai</i>	Chow et al. (2010)
14	dehydrosimplexin	<i>P. elongate, P. nisidai</i>	Chow et al. (2010)
15	acetoxysimplexin	<i>P. elongata, P. nisidai</i>	Chow et al. (2010)
16	gniditrin	<i>D. mezereum</i>	Gorick and Melzig (2013)
17	YH-12	<i>W. indica</i>	Feng et al. (2018)
18	yuanhuafin	<i>D. genkwa</i>	Chen et al. (2013), Zhan et al. (2005)
19	genkwadaphnin	<i>D. genkwa</i>	Bailly (2022)
20	daphnetoxin	<i>D. mezereum</i>	Diogo et al. (2009)
21	YH-15	<i>W. indica</i>	Feng et al. (2018)
22	yuanhuapin	<i>D. genkwa</i>	Chen et al. (2013), Zhan et al. (2005)
23	genkwanine D	<i>D. genkwa</i>	Zhan et al. (2005)
24	genkwanine H	<i>D. genkwa</i>	Zhan et al. (2005)
25	genkwanine A	<i>D. genkwa</i>	Zhan et al. (2005)
26	YH-10	<i>W. indica</i>	Feng et al. (2018)
27	genkwanine J	<i>D. genkwa</i>	Zhan et al. (2005)
28	phorbol-12-benzoate-13-decadienoate	<i>W. canescens</i>	Dagang et al. (1993)
29	prostratin	<i>P. prostrata</i>	Cashmore et al. (1976), Zayed et al. (1977)
30	12-deoxy-5 β -hydroxy-13-tetradecanoylphorbol-6 α ,7 α -oxide	<i>P. simplex</i>	Zayed et al. (1982, 1977)
31	desacetylfevicordin A	<i>P. macrocarpa</i>	Kurnia et al. (2008)
32	fevicordin A	<i>P. macrocarpa</i>	Kurnia et al. (2008)

Table 2 (continued)

No.	Compound	Species	References
33	fevicordin A glucoside	<i>P. macrocarpa</i>	Kurnia et al. (2008)
34	fevicordin D glucoside	<i>P. macrocarpa</i>	Kurnia et al. (2008)

damage, and pupil constriction (Zhao et al., 2004). In clinical practice, *W. chamaedaphne* is used for inducing labor. The expulsion of stillborn fetuses and placental pathological observations indicated its toxic effects on the fetus (Yu et al., 1997).

Contact with the sap of *D. mezereum* could lead to skin rashes, while oral consumption of its plant preparations could cause gastrointestinal disorders (Gorick and Melzig, 2013). Application of fresh *D. palustris* bark on the skin resulted in redness, vascular blockage, and ulcers, while ingestion could lead to severe vomiting and diarrhea (Ramsewak et al., 2001). *P. nisidai* can enhance the body's resistance to external physical, chemical, and biological factors, making it known as a miraculous remedy among Palauans. However, frequent consumption could lead to non-specific complications, diarrhea, and even adverse reactions like death (Kulakowski et al., 2014, 2015).

4.4. Other toxicities

Yan et al. (2014) isolated eight flavonoid compounds from *S. chamaejasme*, which exhibited phytotoxicity by inhibiting the growth of *Arabidopsis thaliana* seedlings. Among them, neochamaejasmin B, daphnodorin, and dihydrodaphnodorin B (45, 48, 49) demonstrated potent phytotoxic activity with IC₅₀ values of 6.9, 7.1, and 27.3 μ g/mL, respectively.

5. Toxicity mechanisms of toxic Traditional Herbals from thymelaeaceae

5.1. DNA strand breakage

Ren et al. (2021) indicated that in *S. chamaejasme*, the compound ICM induced a dose-dependent upregulation of mitochondrial Bax/Bcl-2 expression in AW1 cells, leading to an increased release of cytochrome c (cyt-c) in the cytoplasm. Cyt-c activates caspase 9, which in turn activates caspase 3 (Johnson et al., 2000). Caspase 3 cleaves DNA repair-related protein PARP into DNA fragments, causing cell arrest in the G2 phase. Therefore, the increased expression of the DNA double-strand break marker γ H2AX28 in the cell nucleus suggested that the toxic mechanism of ICM might be associated with DNA double-strand breaks.

The toxicity mechanism of Neochamaejasmin A (NCA, 47) on the *Spodoptera frugiperda* (Sf 9) cell line is similar to that of the ICM in AW1 cells. Detection of PARP cleavage and G2/M arrest in the Sf 9 cell line suggested that the toxicity mechanism of NCA might be associated with DNA double-strand breaks (Ren et al., 2019).

5.2. Oxidative stress

Gstp2 can mitigate the toxicity of endogenous and exogenous substances (Glisic et al., 2015), and a decrease in its expression level suggests a decline in the body's detoxification capacity of antioxidants. SOD1 is an antioxidant enzyme that maintains intracellular ROS homeostasis (Li et al., 2018; Yin et al., 2017). Han et al. (2020) demonstrated that the water extract of *D. genkwa* induced downregulation of Gstp2 and SOD1 gene expression in adult zebrafish liver, resulting in increased ROS levels within the body, leading to oxidative stress damage.

Qi et al. (2015) used ROS fluorescence probes to detect the impact of *D. genkwa* and *W. indica* on ROS levels in Bhas42 cells. They found that

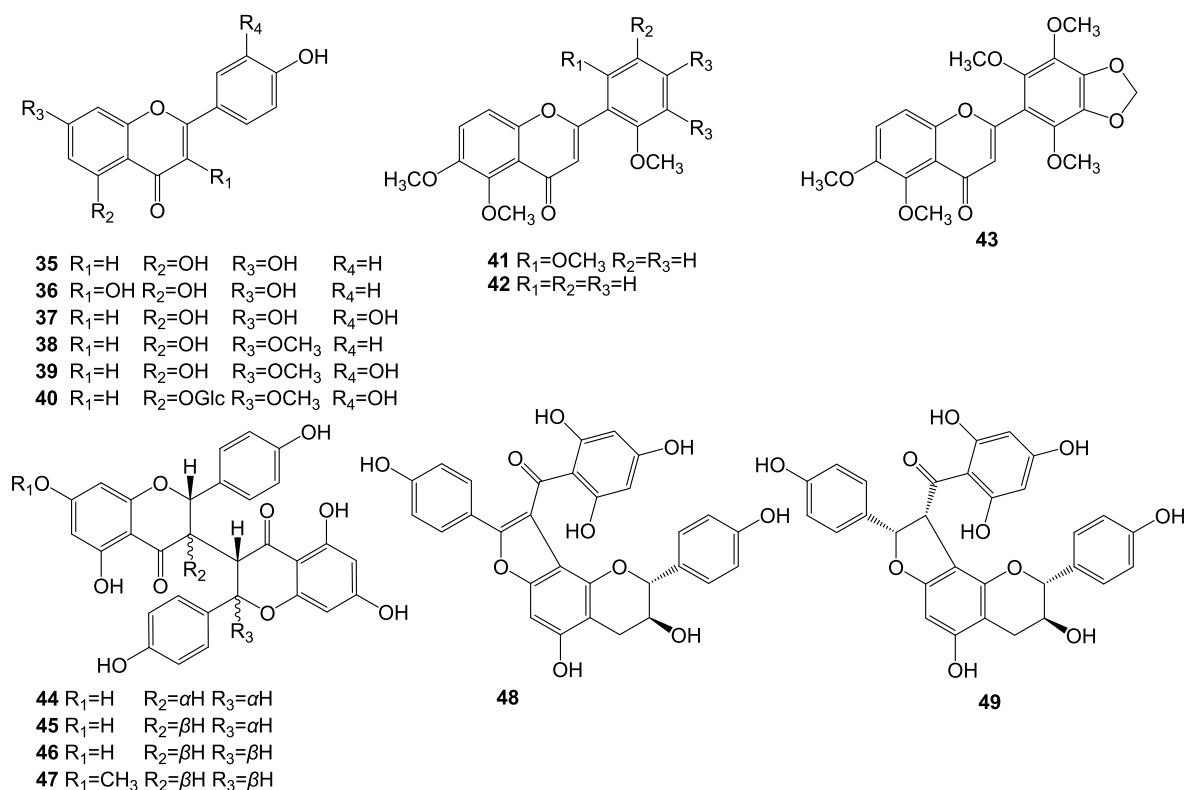


Fig. 2. Toxic flavonoid compounds in toxic Traditional Herbals from Thymelaeaceae.

Table 3

Toxic flavonoid compounds in toxic Traditional Herbals from Thymelaeaceae.

No.	Name	Origins	References
35	apigenin	<i>D. genkwa</i>	Li (2013), Wang et al. (2018)
36	kaempferol	<i>D. genkwa</i>	Li et al. (2007), Wang et al. (2018)
37	luteolin	<i>D. genkwa</i>	Li (2013), Wang et al. (2018)
38	genkwanin	<i>D. genkwa</i>	Li et al. (2007), Wang et al. (2018)
39	3'-hydroxygenkwanin	<i>D. genkwa</i>	Li et al. (2007), Wang et al. (2018)
40	3'-hydroxygenkwanin-5-O- β -D-glucoside	<i>D. genkwa</i>	Wang et al. (2018)
41	5,6,2',6'-tetramethoxyflavone	<i>S. argentea</i>	Ayers et al. (2008)
42	5,6,2'-trimethoxyflavone	<i>S. argentea</i>	Ayers et al. (2008)
43	5,6,2',5',6'-pentamethoxy-3',4'-methylenedioxyflavone	<i>S. argentea</i>	Ayers et al. (2008)
44	isochamaejasmin	<i>S. chamaejasme</i>	Gu et al. (2022)
45	neochamaejasmin B	<i>S. chamaejasme</i>	Gu et al. (2022)
46	neochamaejasmin A	<i>S. chamaejasme</i>	Gu et al. (2022)
47	7-methoxyneochamaejasmin A	<i>S. chamaejasme</i>	Yan et al. (2014)
48	daphnodorin B	<i>S. chamaejasme</i>	Yan et al. (2014)
49	dihydrodaphnodorin B	<i>S. chamaejasme</i>	Yan et al. (2014)

extracts of *D. genkwa* and *W. indica* could dose-dependently elevate ROS levels, suggesting that oxidative stress damage might be one of the mechanisms underlying the carcinogenic toxicity of *D. genkwa* and *W. indica*. Furthermore, oxidative stress can induce DNA oxidative damage (Marin and Țăranu, 2012). Ren et al. (2021) observed oxidative stress in AW1 cells treated with ICM, leading to an increased expression of the DNA oxidative damage repair protein OGG1.

5.3. Endoplasmic reticulum stress

Han et al. (2020), and Zhao et al. (2021) induced pathological

changes in zebrafish liver cells using *D. genkwa* aqueous extract. Q-RT-PCR was used to measure the mRNA expression levels of endoplasmic reticulum stress-related genes *atf4a* and *hsp90b1*, which were significantly upregulated in zebrafish. This suggested that endoplasmic reticulum stress might be one of the mechanisms underlying *D. genkwa*-induced liver damage in zebrafish.

5.4. Impaired tricarboxylic acid cycle (TCA cycle)

Succinate dehydrogenase (SDH) is a critical enzyme in cellular aerobic respiration, and its activity serves as an indicator to evaluate the extent of the TCA cycle. *D. genkwa* induced a significant decrease ($P < 0.05$) in SDH content in the livers of adult zebrafish and SD rats (Han et al., 2020). Its water extract could induce the downregulation of the TCA cycle-related gene *pdha 1a* in zebrafish liver cells, affecting the TCA cycle and resulting in mitochondrial dysfunction (Zhao et al., 2021).

5.5. Lipid metabolism disorder

The compositional element of lipids, triglycerides (TG), maintains a dynamic equilibrium under normal circumstances, and an elevation in its content can indicate disrupted lipid metabolism in liver cells. Han et al. (2020) demonstrated that, compared to the control group, administration of *D. genkwa* water extract significantly increased TG levels in SD rats ($P < 0.01$), indicating that *D. genkwa* water extract induced lipid metabolism disorders, consequently leading to liver damage.

5.6. Alterations in enzyme activity

Zhao (2004) revealed the toxic mechanism of *W. chamaedaphne* on *Tetranychus viennensis* from the perspective of enzyme activity. The active fraction of *W. chamaedaphne* acted on *T. viennensis*, causing significant and rapid changes in detoxification enzymes and active target enzymes within *T. viennensis*. This severely inhibited peripheral neural

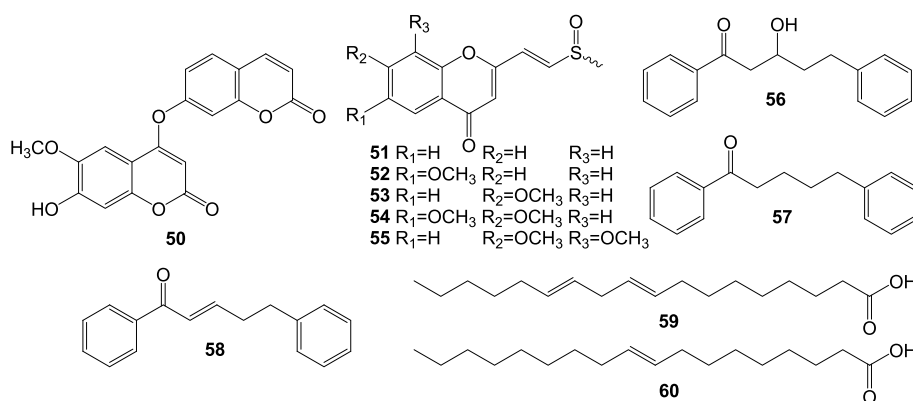


Fig. 3. Other toxic components in toxic Traditional Herbs from Thymelaeaceae.

Table 4
Other toxic components in toxic Traditional Herbs from Thymelaeaceae.

No.	Name	Origins	References
50	isodaphnoretin	<i>D. genkwa</i>	Wang et al. (2018)
51	dirchromone	<i>D. palustris</i>	St-Gelais et al. (2015)
52	6-methoxydirchromone	<i>D. palustris</i>	St-Gelais et al. (2015)
53	7-methoxydirchromone	<i>D. palustris</i>	St-Gelais et al. (2015)
54	6,7-dimethoxydirchromone	<i>D. palustris</i>	St-Gelais et al. (2015)
55	7,8-dimethoxydirchromone	<i>D. palustris</i>	St-Gelais et al. (2015)
56	1,5-diphenyl-3-hydroxyl-1-pentanone	<i>S. chamaejasme</i>	He et al. (2006)
57	1,5-diphenyl-1-pentanone	<i>S. chamaejasme</i>	Gao et al. (2001)
58	1,5-diphenyl-2-penten-1-one	<i>S. chamaejasme</i>	Gao et al. (2001)
59	linoleic acid	<i>D. palustris</i>	Ramsewak et al. (2001)
60	oleic acid	<i>D. palustris</i>	Ramsewak et al. (2001)

synaptic transmission, leading to convulsive symptoms and eventual exhaustion and death.

Gao et al. (2022) discovered that the root extract of *S. chamaejasme* exhibited insecticidal activity against *F. esculentum*, and this activity was

associated with changes in the activities of key enzymes within the insects' bodies, including target enzymes such as acetylcholinesterase and adenosine triphosphatase, as well as detoxifying enzymes like glutathione *S*-transferase, cytochrome P450, and carboxylesterase.

The toxicity mechanisms of toxic Traditional Herbs from Thymelaeaceae are illustrated in Fig. 5. Generally, toxic Traditional Herbs possess potent pharmacological activities (Liu et al., 2017), and studying their toxicity mechanisms contributes to their better utilization in disease treatment. The toxicity mechanisms of many toxic Traditional Herbs from Thymelaeaceae remain unexplored. Investigating their toxicity mechanisms and predicting their toxicities through integrating network toxicology and metabolomics holds significant importance (Li et al., 2019, 2023).

6. Detoxification methods of toxic Traditional Herbs from thymelaeaceae

6.1. Vinegar processing detoxification method

The vinegar-processing method of *D. genkwa* can achieve the goal of reducing toxicity and moderating medicinal properties. This can be achieved by the following procedure: First, take clean *D. genkwa*, add a specific amount of rice vinegar, mix thoroughly, and let it absorb the vinegar until dry. Then, heat it in a frying container over a gentle flame until slightly dry. Remove and let it cool (Liu et al., 2019). Using this method, Wang and Cao (1999) detected a significant decrease in the

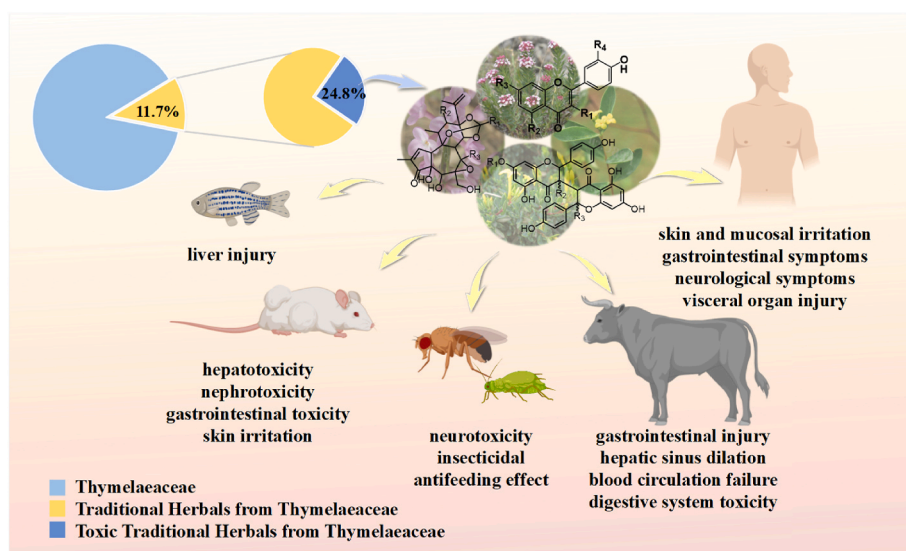


Fig. 4. Toxicity of toxic Traditional Herbs from Thymelaeaceae to animals and humans.

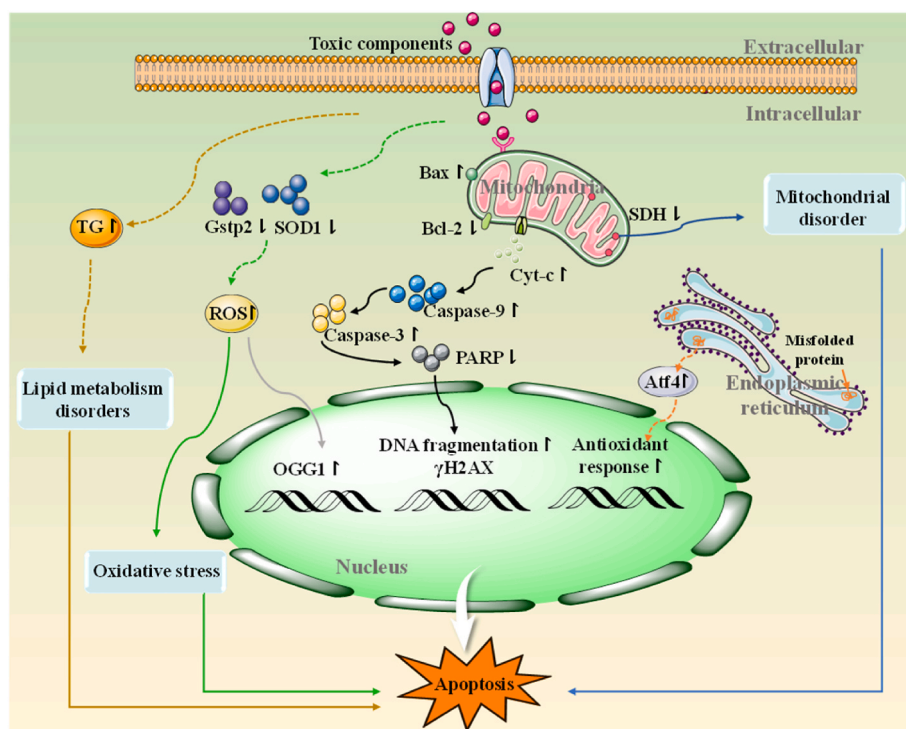


Fig. 5. The toxicity mechanisms of toxic Traditional Herbs from Thymelaeaceae.

content of yuanhuacine (9) in the vinegar-fried *D. genkwa* compared to other processing methods. During the vinegar-frying process, influenced by acetic acid and heating, glycosides undergo hydrolysis, increasing the content of glycosidic elements. This process enhances the content of cough-suppressing active components such as hydroxycanthinone (39) and cantharidin (38) in *D. genkwa*.

S. chamaejasme in its raw state, exhibits severe toxicity and is rarely ingested orally. Instead, it is predominantly used externally for insecticidal purposes and to treat skin-related ailments. In the "Sheng Hui Fang," *S. chamaejasme* processed with vinegar is used to treat accumulation and abdominal distension. After vinegar-frying, the toxicity of *S. chamaejasme* is reduced, making it suitable for internal use. The vinegar-frying method for *S. chamaejasme* is similar to that of *D. genkwa* (Liu et al., 2019).

In Unani medicine, it is recommended to choose *D. oleoides* with broader leaves. Soak its leaves in vinegar for three days and three nights, changing the vinegar daily. Afterward, rinse them with water. Subsequently, dry them in a 45 °C oven, grind them into coarse powder. Finally, fry the powder with almond oil to obtain a fine powder. Applying this can detoxify and demonstrate significant anti-inflammatory activity (Rahman et al., 2020).

6.2. Sweat soaking detoxification method

W. indica has demonstrated definite efficacy against certain ailments, but its toxicological research is still incomplete. Overconsumption, raw ingestion, or inadequate brewing time may lead to a series of toxic side effects such as vomiting, diarrhea, chest tightness, and shock. The sweat-soaking method is commonly applied to processing the Miao medicine *W. indica* to reduce toxicity, significantly decreasing its acute and sub-acute toxicity while maintaining its inherent efficacy. Take 5 kg of coarse powder of raw *W. indica* and spray it with 1% artificial sweat (containing 0.5 g/L L-ammonium salt, 5 g/L sodium chloride, and 2.2 g/L disodium hydrogen phosphate dihydrate) each day. Adjust the pH to 5.5 with 0.05 mol/L sodium hydroxide, mix well, let it soak, and once saturated, place it in an oven at 37 °C for 24 h. For every 100 kg of

W. indica requires 30 kg of artificial sweat, and this process is repeated for 14 days. (Peng et al., 2022). The analgesic and anti-inflammatory effects of raw and processed *W. indica* show no significant difference. However, the processed *W. indica* exhibits significantly reduced toxicity in terms of liver and kidney damage. Three diterpenoid compounds found in *W. indica*, YH-10 (26), YH-12 (17), and YH-15 (21), have been proven to be highly toxic. Feng et al. discovered that after sweat processing of *W. indica*, the content of these three diterpenoid compounds decreased by 48%, 44%, and 65%, respectively (Feng et al., 2018).

6.3. Rational compatibility detoxification method

In Zhang Zhihe's "Confucians' Duties to Their Parents," the "anti-drug compatibility" records that the concurrent use of *D. genkwa* and *Glycyrrhiza uralensis* not only diminishes each other's efficacy but can also lead to severe toxicity or adverse effects. Chen et al. (2019) evaluated the incompatibility of the combination of *D. genkwa* and *G. uralensis*: Rats exposed to a dosage of 1.0 g/kg *D. genkwa* (three times the maximum dosage specified in the Chinese Pharmacopoeia) exhibited toxicity to the heart, liver, kidneys, and testes. Rats exposed to a combination of (1.0 g/kg *D. genkwa* + 1.0 g/kg *G. uralensis*) showed similar liver toxicity and exacerbated kidney and reproductive toxicity. There are limited reports on the solo use of *S. chamaejasme*. In clinical practice, it is often used in combination with other drugs. For instance, "Taiping Shenghui Fang" records the use of a formulation containing *S. chamaejasme*, *D. genkwa*, and *Aconitum carmichaelii* for treating conditions such as fatigue and abdominal distension (Zhang et al., 2021).

6.4. Dose control detoxification method

The dosage and duration of drug use are also critical factors affecting both efficacy and safety, particularly for Traditional Herbs containing toxic components. Overdosage and prolonged use of such substances should be inhibited to prevent potential risks (Bailly, 2022; Chen et al., 2013). The 2020 edition of the "Chinese Pharmacopoeia" provides specific information regarding the recommended daily dosage and single

dose for *D. genkwa*. For vinegar-processed *D. genkwa*, the suggested oral dosage is 0.6–0.9 g per administration, taken once a day. Another example is *S. chamaejasme*, which is recorded only for external use in appropriate amounts (Meng et al., 2023).

6.5. Other detoxification methods

Due to the higher concentration of toxic diterpenoid esters in the methanol extract of *P. nissidai*, the water extraction method can avoid the extraction of such toxic components. Therefore, water extraction is the traditional preparation method for *P. nissidai*, preserving its immunostimulatory activity while reducing toxicity (Kulakowski et al., 2015). In Tibetan medicine, there are records of detoxification methods for *S. chamaejasme*, including milk processing, wine boiling, boiling with child's urine, and boiling with lamb soup (Gao and Xu, 1999).

7. Conclusion and outlook

Toxic Traditional Herbals from Thymelaeaceae are widely used to treat conditions such as HIV/AIDS, leukemia, skin disorders, trauma, and insecticide application. However, their toxicity limits their clinical application, leading to adverse reactions such as organ damage, gastrointestinal symptoms, and skin and mucosal irritations when improperly used. The toxicity can be mitigated through proper processing, rational compatibility, and adhering to principles of proper usage and dosage, thereby minimizing the potential adverse effects.

Currently, there are 60 reported toxic components found in toxic Traditional Herbals from Thymelaeaceae. Some of these components exhibit significant pharmacological activities along with certain toxicity levels, such as diterpenoids and flavonoids. The mechanisms underlying their toxicity are related to DNA damage, oxidative stress, endoplasmic reticulum stress, and inhibition of the tricarboxylic acid cycle. "Vinegar-processing" and "sweat-soaking" are recognized detoxification methods for *D. genkwa* and *W. indica*, respectively, in the medical field. However, detoxification methods for other toxic medicinal plants within this family still need further exploration and verification.

In recent years, medicinal plants from the Thymelaeaceae family have gained significant attention due to their diverse pharmacological activities and promising applications. Evaluating and summarizing their toxicity is beneficial for the development and clinical treatment of these medicinal plants. Currently, the reported toxicity studies of Thymelaeaceae medicinal plants primarily focus on in vitro and animal experiments, with limited documentation on clinical toxicity. Reports on plants like *W. indica* and *S. chamaejasme* have indicated potential toxic plant parts, but the specific toxic compounds underlying this toxicity require further investigation. Initial speculation on possible toxic mechanisms, such as those of *W. chamaedaphne*, has been based on changes in gene expression levels and enzyme activity. However, a more detailed exploration is needed to fully understand the specific processes leading to toxicity. In summary, medicinal plants from the Thymelaeaceae family exhibit both pharmacological activities and toxicities. A comprehensive understanding of their toxicity and potential mechanisms is crucial for toxicity control and enhancing their therapeutic effects. This will ultimately contribute to the improved utilization of Thymelaeaceae medicinal plants for treating human diseases.

CRediT authorship contribution statement

Lingnan Zhao: Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Yanping Zhang:** Investigation, Methodology, Conceptualization. **Qianqian Yin:** Conceptualization, Software. **Gang Chen:** Data curation, Project administration. **Wei Li:** Data curation, Project administration. **Ning Li:** Data curation, Funding acquisition, Methodology, Project administration, Supervision.

Declaration of competing interest

The authors declare no conflict of interest pertaining to this manuscript.

Data availability

The data that has been used is confidential.

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Abbreviations

AIDS	acquired immunodeficiency syndrome
CNKI	China national knowledge infrastructure
HIV	human immunodeficiency virus
ICM	isochamaejasmin
ROS	reactive oxygen species
NCB	neochamaejasmin B
ALP	alkaline phosphatase
AI	antifeedant indice
Cyt-c	cytochrome c
PARP	poly ADP-ribose polymerase
Sf	9 <i>Spodoptera frugiperda</i>
SOD1	superoxide dismutase 1
PCR	polymerase chain reaction
q-RT-PCR	quantitative real-time PCR
SDH	succinate dehydrogenase
TCA	tricarboxylic acid cycle
TG	triglyceride
SD	Sprague Dawley
T-BIL	total bilirubin
GLU	glutamic acid
A/G	albumin-globulin ratio
TP	total protein
ALB	albumin

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